Magnetic field of the two-dimensional loop

1 Problem Setup

The conduction calculation gives an in-plane current $\boldsymbol{J} = (J_x, J_y, 0)$. The model is unchanged along the z direction and assumes uniform permeability μ, with no applied magnetic field. Under these assumptions, the magnetic field has only a z component: $\boldsymbol{B} = (0, 0, B_z(x, y))$.

Ampere's law, $\nabla \times \boldsymbol{B} = \mu \boldsymbol{J}$, then becomes

$$(\partial_y B_z, -\partial_x B_z) = \mu(J_x, J_y), \qquad \nabla B_z = \mu(-J_y, J_x). \tag{1}$$

There is no current in air, so $\nabla B_z = 0$ there: B_z is constant in each connected air region. The outside constant is set to zero. The cavity constant, called C_B, must be determined from the current. With no idealized sheet current at the conductor surfaces, B_z is continuous. Thus it is zero on the outer conductor surface and C_B on the cavity-facing surface.

2 Fit the B_z field

Equation (1) suggests integrating $\mu(-J_y, J_x)$ along a path to obtain B_z. However, due to numeric errors of the finite-element method, different paths could give different approximate values. This motivates an optimization approach below.

Let $\boldsymbol{g}_{*,h} = \mu(-J_{y,h}, J_{x,h})$ be the gradient requested by Ampere's law in conductor triangle h. Represent B_z by a continuous piecewise-linear function $\widetilde{B}_z = \sum_i b_i N_i$, where the summation is over all conductor interior vertices and conductor-cavity vertices. We further constraint all conductor-cavity vertices share the same $b_i = C_B$. With this choice of vertices, the boundary conditions are automatically satisfied, that is,

$$\widetilde{B}_z|_{\Gamma_{\text{outer}}} = 0, \quad \widetilde{B}_z|_{\Gamma_{\text{cavity}}} = C_B. \tag{2}$$

Our goal is to minimize the area-weighted difference:

$$\min_{\widetilde{B}_z} \frac{1}{2} \sum_{h \in \mathcal{T}_c} A_h \left| \sum_{i=0}^{2} b_{h,i} \nabla N_{h,i} - \boldsymbol{g}_{*,h} \right|^2, \tag{3}$$

Here $\mathcal{T}_c$ is the set of conductor triangles, and $b_{h,i}$ is the value of $\widetilde{B}_z$ at local vertex i of triangle h. It is understood that $b_{h,i} = 0$ if the vertex falls onto conductor-outer boundary and $b_{h,i} = C_B$ if the vertex falls onto conductor-cavity boundary. Taking derivative to Equation 3 gives a linear system that we can solve to get $\widetilde{B}_z$.

3 Field used by the app

The app stores the solved B_z at conductor vertices, fills cavity vertices with C_B, and fills outside vertices with zero. The magnetic view interpolates those nodal values within each triangle. The energy-flow calculation uses the triangle average

$$\overline{B}_{z,h} = \frac{b_{h,0} + b_{h,1} + b_{h,2}}{3}. \tag{4}$$

This is a strictly two-dimensional magnetic model; it is not the magnetic field of a finite three-dimensional wire.

Code: `calculations/magnetics.py`.